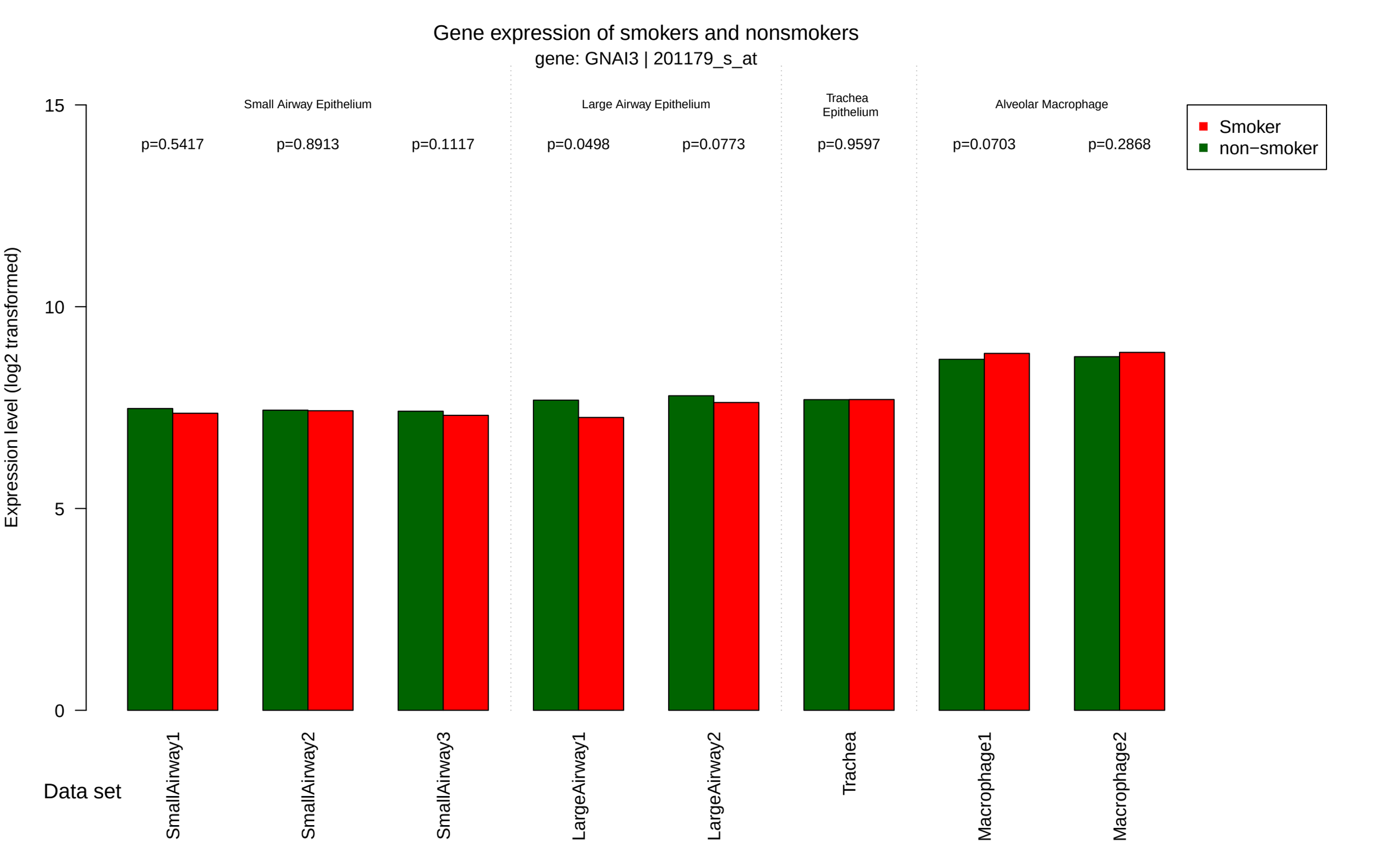
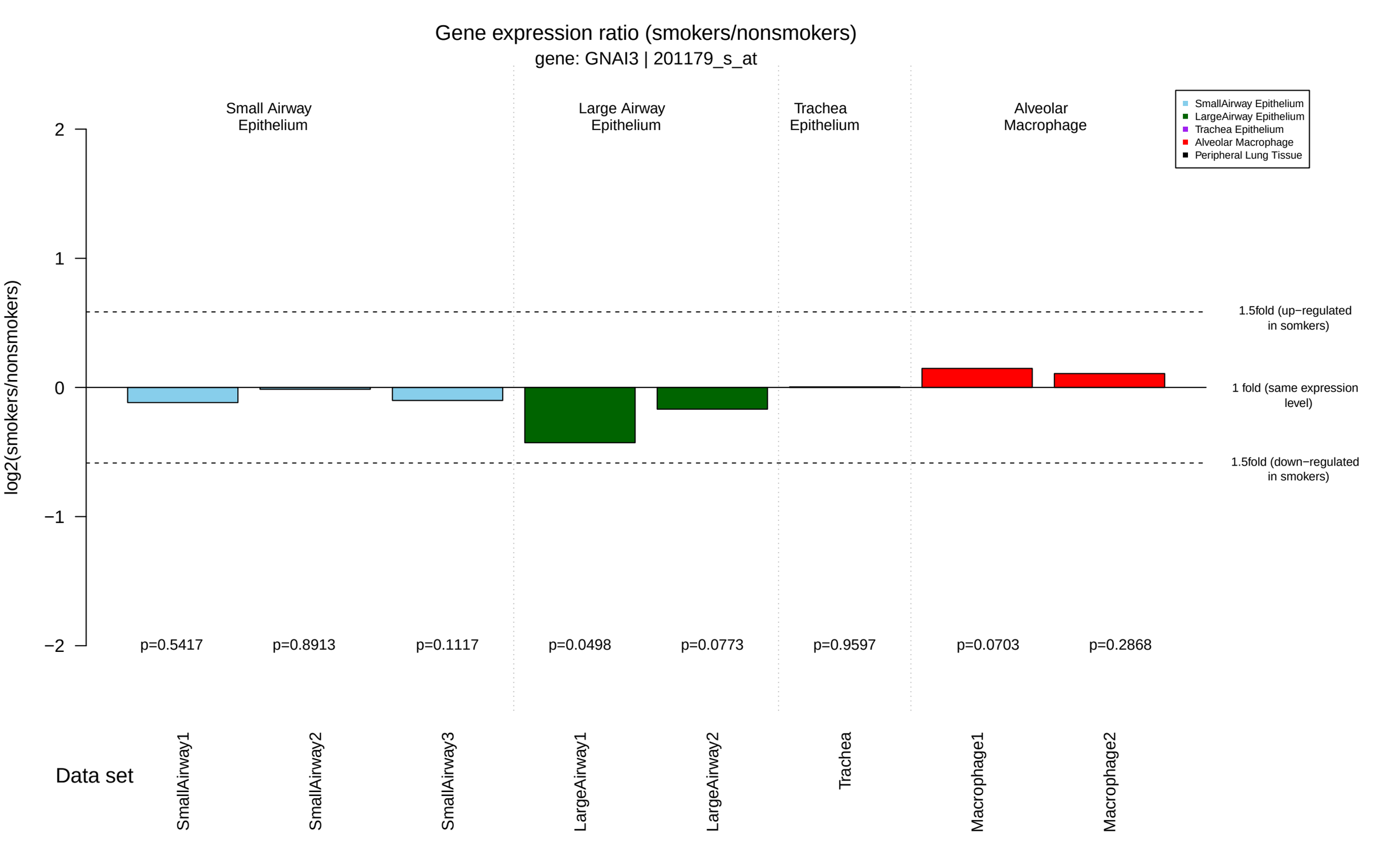
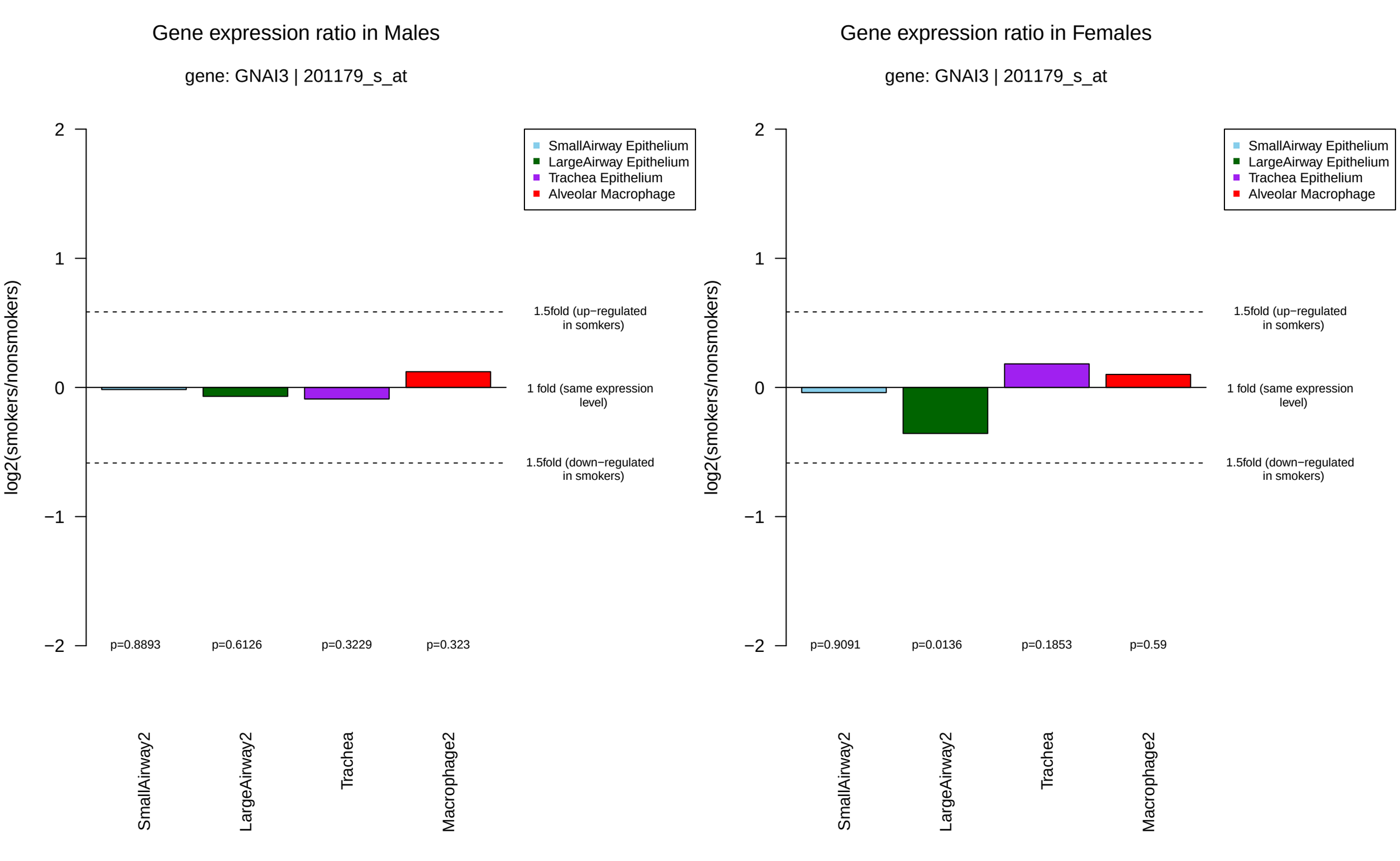
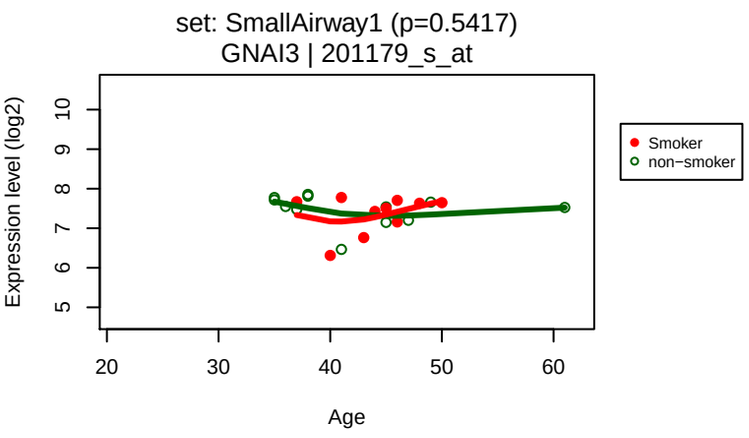
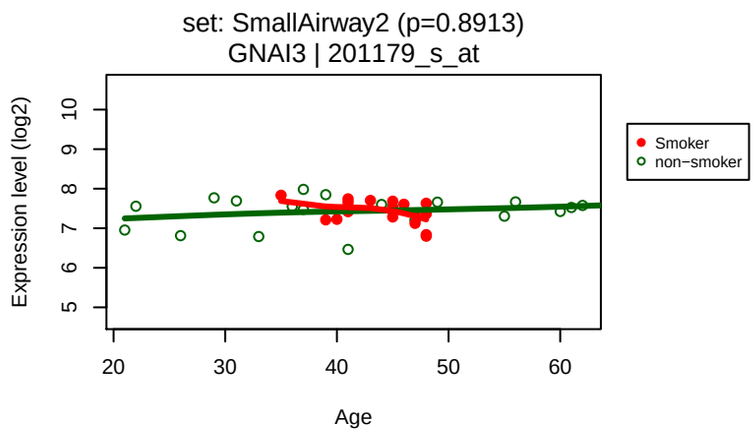
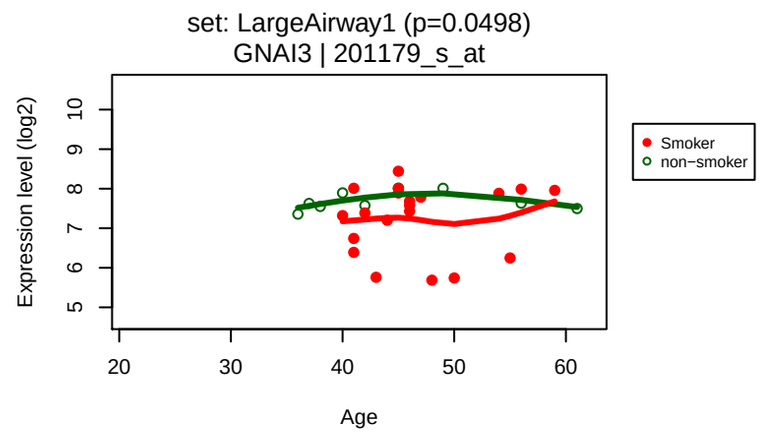

Gene expression of smokers and nonsmokers
gene: GNAI3 | 201179_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5417
p=0.8913
p=0.1117
p=0.0498
p=0.0773
p=0.9597
p=0.0703
p=0.2868
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

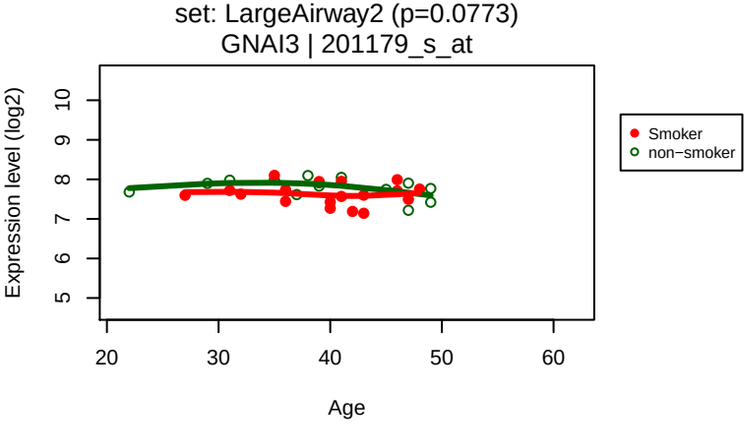

Gene expression ratio (smokers/nonsmokers)
gene: GNAI3 | 201179_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5417
p=0.8913
p=0.1117
p=0.0498
p=0.0773
p=0.9597
p=0.0703
p=0.2868
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

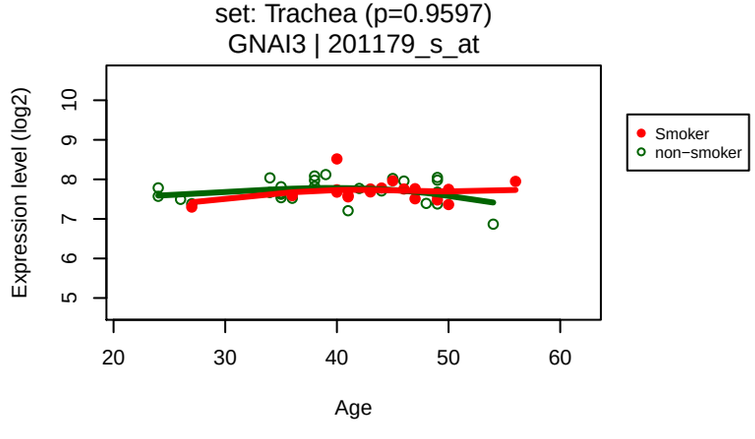

Gene expression ratio in Males
gene: GNAI3 | 201179_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8893
p=0.6126
p=0.3229
p=0.323
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNAI3 | 201179_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9091
p=0.0136
p=0.1853
p=0.59
SmallAirway2
LargeAirway2
Trachea
Macrophage2

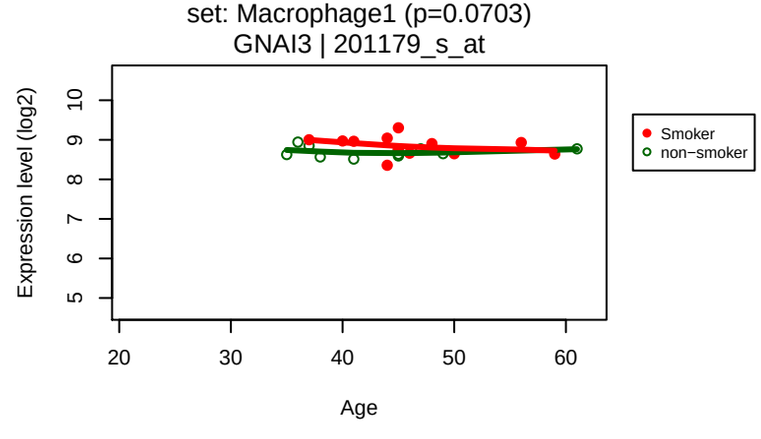

set: SmallAirway1 (p=0.5417)
GNAI3 | 201179_s_at
Expression level (log2)
Age
Smoker
non−smoker

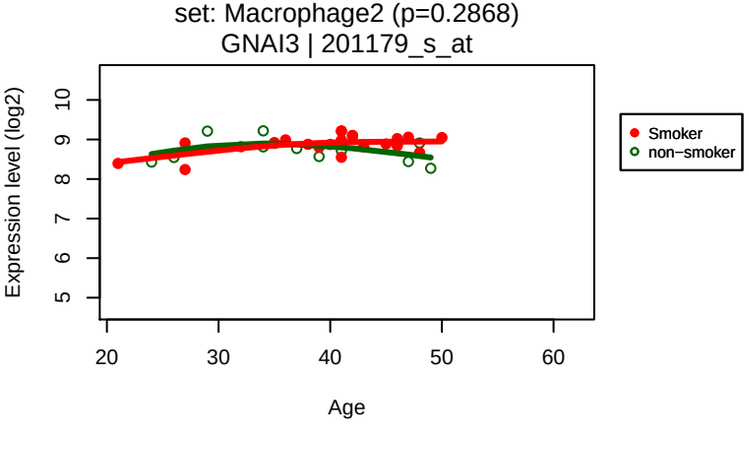

set: SmallAirway2 (p=0.8913)
GNAI3 | 201179_s_at
Expression level (log2)
Age
Smoker
non−smoker

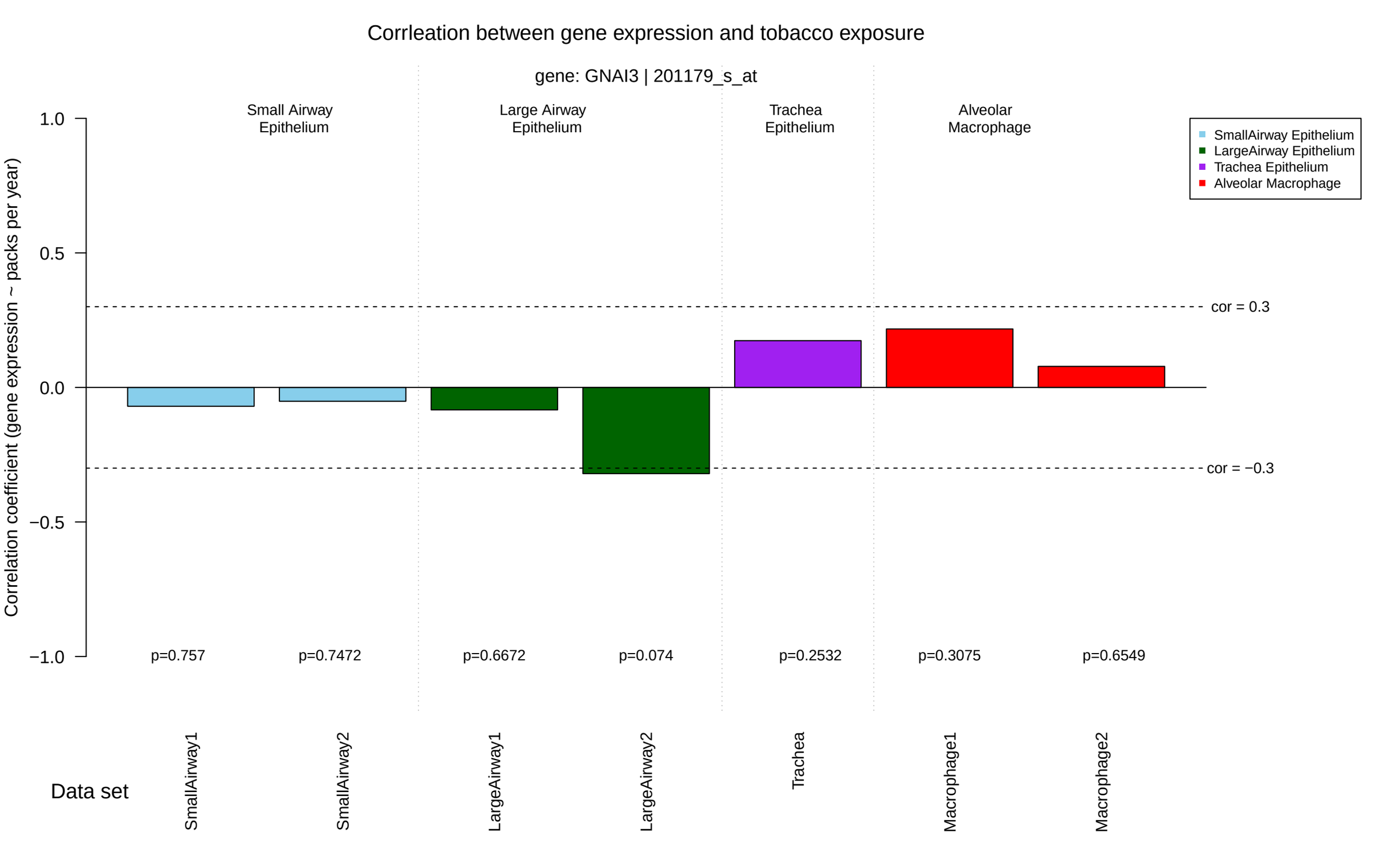

set: LargeAirway1 (p=0.0498)
GNAI3 | 201179_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0773)
GNAI3 | 201179_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9597)
GNAI3 | 201179_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0703)
GNAI3 | 201179_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2868)
GNAI3 | 201179_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNAI3 | 201179_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.757
p=0.7472
p=0.6672
p=0.074
p=0.2532
p=0.3075
p=0.6549
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2